Qn. Set Code-7

Semester: 5th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE, E&CSE

AUTUMN END SEMESTER EXAMINATION-2024

5th Semester B.Tech

DESIGN AND ANALYSIS OF ALGORITHMS

CS 30001 / CS 2012

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Arrange the following in increasing order of Time Complexity

$f1 = 2^n$, $f2 = n^{\log n}$, $f3 = n^{\sqrt{n}}$, $f4 = n^2$

(b) Apply the master method to determine the asymptotic behavior of the function T(n).

$T(n) = (\frac{1}{2})\, T(n/2) + 1/n$

(c) Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are TRUE?

I. Quicksort runs in $\Theta(n^2)$ time

II. Bubblesort runs in $\Theta(n^2)$ time

III. Mergesort runs in $\Theta(n)$ time

IV. Insertion sort runs in $\Theta(n)$ time

a) I and II only

b) I and III only

c) II and IV only

d) I and IV only

(d) Let the frequency of characters i, n, d, e, x be 16, 7, 17, 25, 20 respectively. What would be the Huffman code for the character x?

(e) Consider the following function:

```
int fun(int n)
{
  int i, j, k = 0;
  for (i = 1; i <= n; i++)
    for (j = 1; j <= n; j = j * 2)
        k = k + 1;
    return k;
}
```

What is the execution time of the above function?

a) $\Theta(\log n)$

b) $\Theta(n \log n)$

c) $\Theta(n)$

d) $\Theta(n^2)$

(f) A graph has n vertices. By how much will the total weight of the spanning tree be increased, if the weight of each edge of the graph is increased by 3 units?

(g) Which method is adopted in LCS problem?

a) Divide & Conquer Strategy

b) Back Tracking

c) Greedy Approach

d) Dynamic Programming

(h) Which of the following methods is not used for amortized analysis?

a) Accounting method

b) Potential function method

c) Sorting method

d) Aggregate method

(i) Differentiate between Las Vegas and Monte Carlo algorithms with examples.

(j) Shahrukh and Salman have been asked to show that a certain problem "Superstar" is NP-complete. Shahrukh shows a polynomial time reduction from the 3-SAT problem to "Superstar" and Salman shows a polynomial time reduction from "Superstar" to 3-SAT. Which of the following can be inferred from these reductions?

a) "Superstar" is NP-hard but not NP-complete

b) "Superstar" is NP-complete

c) "Superstar" is in NP, but is not NP-complete

d) "Superstar" is neither NP-hard, nor in NP

SECTION-B

2. (a) Solve the following recurrence relations [5]

(i) $T(n) = 2T(\sqrt{n}) + 1$; for $n > 2$

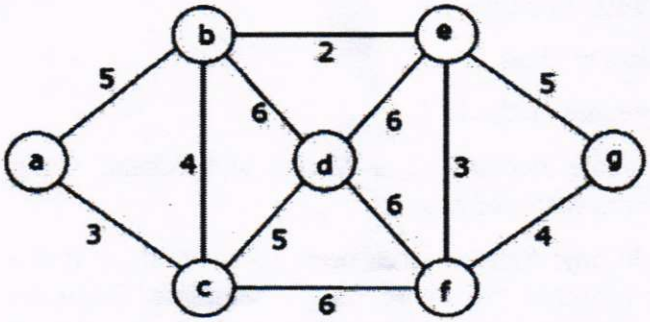
$T(2) = 1$

(ii) $T(n) = 2T(n/2) + 4n$; for $n>1$

$T(1) = 1$

(b) Show the application of PARTITION() procedure at each step on the array A = {98, 87, 76, 65, 54, 43, 32, 21, 10}. Determine the recurrence relation and calculate the time complexity in order to partition this array. [5]

3. (a) Show the contents of the disjoint set data structure when cycle is formed during constructing the Minimum Cost Spanning Tree using Kruskal's Algorithm for the graph given below. [5]

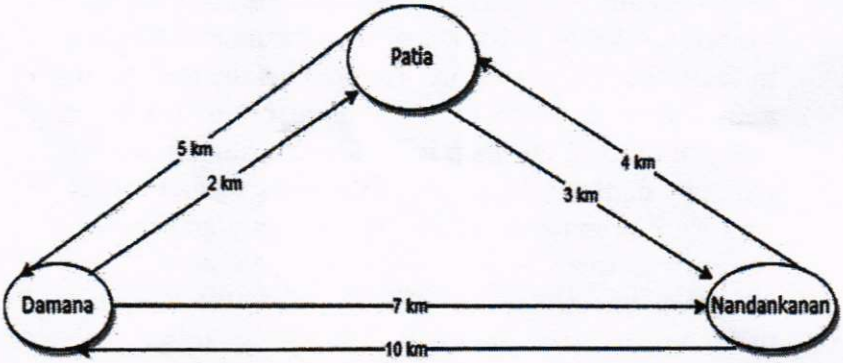


(b) A set of possible activities with rewards earned and it's deadlines are given for the post lunch session on a day in the table below. Organize an optimal sequencing of activities for which a KIITIAN's will get an opportunity to earn maximum rewards? Each activity takes exactly one hour. [5]

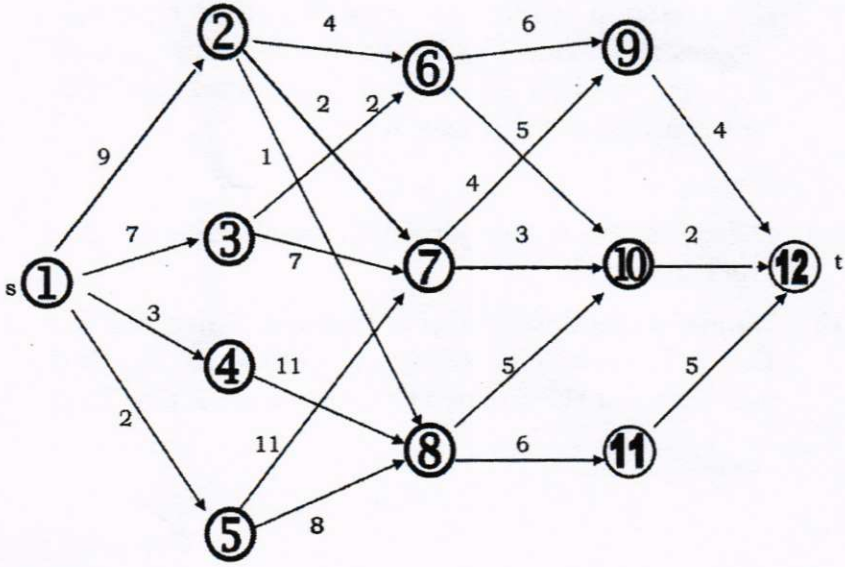
Activities	Deadlines	Reward (Rs)
Meet the Tutor Mentor	By 5 PM	200
Visit the library to study	By 3 PM	180
Meet course faculty	By 3 PM	190
Meet the DG	By 2 PM	300
Activity at KSAC	By 1 PM	120
Meet the Dean	By 2 PM	100

4. (a) Write down an algorithm to store the sum of the elements present in each of the upper diagonal of a two dimensional matrix of size NxN to an array of size N. Find it's execution time. [5]

(b) MBharat LPG Distributors wants to identify the best routes for hassle free delivery between Patia, Damana and Nandankanan. They have obtained a geographical map of these areas along with the distance. Plan the best route between the three areas so that each area can be reached quickly from the other for providing better customer service. [5]



5. (a) Find the minimum cost path to go to "t" from "s". What will be the Time complexity of this problem? [5]



(b) Suppose we perform a sequence of n operations on a data structure in which the ith operation costs i if i is an exact power of 2, and 1 otherwise. Use aggregate analysis or accounting method to determine the amortized cost per operation. [5]

6. (a) Syambahadur is a smart shopkeeper. He uses an array simply to keep a track of the customer purchase. Initially the array is empty. He doubles the size of this array whenever there is no more space in it. Cost of every product in his shop is Rs 5 and costumer can buy only one item at a time, out of those Rs 5, shopkeepers uses Rs 2 to make an entry into the array and reserves Rs 3 in a fund to copy previous data in the array whenever the size is doubled. Calculate the amortized time complexity by showing a detailed working. [5]

(b) In the campus recruitment process Ramesh and Suresh are asked to write an algorithm to sort an array using Quicksort. Ramesh writes an algorithm that always chooses the last element as the pivot to partition the array. Suresh writes an algorithm that arbitrarily chooses any element to be the pivot. Evaluate whose algorithm will have a lesser worst case time complexity with substancial mathematical proof. [5]

7. (a) Define NP. Write an algorithm to proof Clique decision problem is in NP. [5]

(b) Define a Hamiltonian Cycle Problem. Prove that the Decision Travelling Salesman Problem is in NPC assuming that Hamiltonian Cycle Problem is in NPC. [5]

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